

Ishaan Rai

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SUMMARY

Computer Science Engineering student (2027, CGPA 8.17) with strong foundations in data structures, algorithms, OOP, and large-scale distributed systems. Proficient in Java with hands-on experience building scalable, high-performance backend systems in production. Proven ability to architect service-oriented, multi-tiered software, integrate AI/ML capabilities, and deliver in fast-paced team environments. Self-driven and detail-oriented.

EDUCATION

Chitkara University

Bachelor of Technology in Computer Science and Engineering, CGPA: 8.17/10.0 | Dean's List

Punjab, India

Aug 2023 – Sep 2027

TECHNICAL SKILLS

CS Fundamentals: Data structures and algorithms, object-oriented programming and design, complexity analysis, problem solving, system design, operating systems concepts

Languages: Java (primary), Python, C++, JavaScript, TypeScript, SQL, Shell/Bash

Scalable & Distributed Systems: Microservices, Apache Kafka, service-oriented architecture, high-performance and reliable systems, async processing, large-scale distributed computing

Databases: PostgreSQL, MySQL, MongoDB — relational schema design, query optimisation, index-aware data modelling, large-scale data management

AI/ML Integration: LLM API integration, AI-powered feature development, Generative AI, prompt engineering

Cloud & Infrastructure: AWS (EC2, S3, RDS, Lambda, CloudWatch), Docker, Kubernetes, CI/CD, Git

EXPERIENCE

Freelance Software Engineer

Jan 2026 – Present

Self-Employed

Remote

- Designed and built scalable backend systems in Java for 3+ clients operating in fast-paced, results-driven environments; applied object-oriented design, data structures, and complexity analysis to architect reliable, high-performance solutions from broadly defined requirements.
- Engineered RESTful service-oriented integrations with third-party APIs; resolved data pipeline issues through root cause analysis and systematic debugging, delivering reliable software consistently on schedule.

Organizing Head — DevLearn Technical Community

Jan 2024 – Present

Chitkara University

Punjab, India

- Led 10+ structured workshops for 60+ students on data structures, algorithms, OOP, and distributed systems; demonstrated entrepreneurial ownership, strong interpersonal communication, and ability to drive measurable outcomes in a collaborative team environment.

PROJECTS

NeuroSync AI – Large-Scale Distributed System — Java, Apache Kafka, MongoDB — Demo

Nov 2024

- Architected a high-performance, scalable distributed system using Apache Kafka for async event streaming and MongoDB for large-scale data storage with index-optimised query performance; applied service-oriented architecture with independently deployable microservice tiers to ensure fault tolerance and reliability under concurrent load.
- Applied rigorous complexity analysis to message-routing algorithms and data structure selection to meet latency targets at sustained throughput — demonstrating CS fundamentals applied to a real distributed computing problem.

FitTracker – Scalable Production Platform — Java 21, Spring Boot, PostgreSQL, AWS — Live

Jan 2025

- Built and deployed a scalable, reliable Java backend using object-oriented design and normalised relational schema in PostgreSQL; scaled to 300+ active users post-launch with zero critical incidents — demonstrating end-to-end ownership from design through high-quality production delivery.
- Deployed on AWS (EC2, S3, RDS) with Docker; used CloudWatch for real-time monitoring, proactive alerting, and performance tuning under live production load.

Luxe Health AI – Cloud-Native Microservices — Java, Spring Boot, Docker, Kubernetes, AWS — Demo

Jan 2025

- Built and orchestrated fault-tolerant containerised microservices on AWS EC2 using Docker and Kubernetes; enforced independently deployable service boundaries in a multi-tiered distributed architecture with AI/ML feature integration and real-time CloudWatch monitoring.

HackMap India – Community Platform — React, Node.js, PostgreSQL

Oct 2024

- Designed and shipped a full-stack platform serving 200+ users; built a Node.js backend with PostgreSQL, implemented efficient data aggregation queries, and iterated rapidly on features based on user feedback in a self-driven development cycle.

ACHIEVEMENTS

Microsoft Code Cubicles: Top 5 Finalist | Meraki Hackathon: 3rd / 250+ teams | Gamethon: Winner (INR 6K) | TechHacks, HackIndia, Hack5ive: Multiple top placements | Dean's List | Top Project Team Award